

Vantagens:

- 1) para gerar uma string de tamanho N , é necessário aplicar N regras.
- 2) Facilmente convertida em automato de pilha real-time.

1 Greibach Normal Form (GNF)

A CFG $G = (V, T, R, S)$ is said to be in GNF if every production is of the form $A \rightarrow a\alpha$, where $a \in T$ and $\alpha \in V^*$, i.e., α is a string of zero or more variables.

Definition: A production $\mathcal{U} \in R$ is said to be in the form **left recursion**, if $\mathcal{U} : A \rightarrow A\alpha$ for some $A \in V$.

Left recursion in R can be eliminated by the following scheme:

- If $A \rightarrow A\alpha_1 | A\alpha_2 | \dots | A\alpha_r | \beta_1 | \beta_2 | \dots | \beta_s$, then replace the above rules by
- (i) $Z \rightarrow \alpha_i | \alpha_i Z, 1 \leq i \leq r$ and (ii) $A \rightarrow \beta_i | \beta_i Z, 1 \leq i \leq s$
- If $G = (V, T, R, S)$ is a CFG, then we can construct another CFG $G_1 = (V_1, T, R_1, S)$ in **Greibach Normal Form (GNF)** such that $L(G_1) = L(G) - \{\epsilon\}$.

The stepwise algorithm is as follows:

1. Eliminate null productions, unit productions and useless symbols from the grammar G and then construct a $G' = (V', T, R', S)$ in **Chomsky Normal Form (CNF)** generating the language $L(G') = L(G) - \{\epsilon\}$.
2. Rename the variables like $A_1, A_2, \dots, A_n$ starting with $S = A_1$.
3. Modify the rules in R' so that if $A_i \rightarrow A_j\gamma \in R'$ then $j > i$
4. Starting with A_1 and proceeding to A_n this is done as follows:
 - (a) Assume that productions have been modified so that for $1 \leq i \leq k$, $A_i \rightarrow A_j\gamma \in R'$ only if $j > i$
 - (b) If $A_k \rightarrow A_j\gamma$ is a production with $j < k$, generate a new set of productions substituting for the A_j the body of each A_j production.
 - (c) Repeating (b) at most $k - 1$ times we obtain rules of the form $A_k \rightarrow A_p\gamma, p \geq k$
 - (d) Replace rules $A_k \rightarrow A_k\gamma$ by removing left-recursion as stated above.
5. Modify the $A_i \rightarrow A_j\gamma$ for $i = n - 1, n - 2, \dots, 1$ in desired form at the same time change the Z production rules.

Example: Convert the following grammar G into Greibach Normal Form (GNF).

$$\boxed{\begin{array}{l} S \rightarrow XA|BB \\ B \rightarrow b|SB \\ X \rightarrow b \\ A \rightarrow a \end{array}}$$

To write the above grammar G into GNF, we shall follow the following steps:

1. Rewrite G in Chomsky Normal Form (CNF)

It is already in CNF.

2. Re-label the variables

S with A_1

X with A_2

A with A_3

B with A_4

After re-labeling the grammar looks like:

$$A_1 \rightarrow A_2A_3|A_4A_4$$

$$A_4 \rightarrow b|A_1A_4$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

3. Identify all productions which do not conform to any of the types listed below:

$$A_i \rightarrow A_jx_k \text{ such that } j > i$$

$$Z_i \rightarrow A_jx_k \text{ such that } j \leq n$$

$$A_i \rightarrow ax_k \text{ such that } x_k \in V^* \text{ and } a \in T$$

4. $A_4 \rightarrow A_1A_4$ identified

5. $A_4 \rightarrow A_1A_4|b$.

To eliminate A_1 we will use the substitution rule $A_1 \rightarrow A_2A_3|A_4A_4$.

Therefore, we have $A_4 \rightarrow A_2A_3A_4|A_4A_4A_4|b$

The above two productions still do not conform to any of the types in step 3.

Substituting for $A_2 \rightarrow b$

$$A_4 \rightarrow bA_3A_4|A_4A_4A_4|b$$

Now we have to remove left recursive production $A_4 \rightarrow A_4A_4A_4$

$$A_4 \rightarrow bA_3A_4|b|bA_3A_4Z|bZ$$

$$Z \rightarrow A_4A_4|A_4A_4Z$$

6. At this stage our grammar now looks like

$$A_1 \rightarrow A_2A_3|A_4A_4$$

$$A_4 \rightarrow bA_3A_4|b|bA_3A_4Z|bZ$$

$$Z \rightarrow A_4A_4|A_4A_4Z$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

All rules now conform to one of the types in step 3.

But the grammar is still not in Greibach Normal Form!

7. All productions for A_2, A_3 and A_4 are in GNF

for $A_1 \rightarrow A_2A_3|A_4A_4$

Substitute for A_2 and A_4 to convert it to GNF

$$A_1 \rightarrow bA_3|bA_3A_4A_4|bA_4|bA_3A_4ZA_4|bZA_4$$

for $Z \rightarrow A_4A_4|A_4A_4Z$

Substitute for A_4 to convert it to GNF

$$Z \rightarrow bA_3A_4A_4|bA_4|bA_3A_4ZA_4|bZA_4|bA_3A_4A_4Z|bA_4Z|bA_3A_4ZA_4Z|bZA_4Z$$

8. Finally the grammar in GNF is

$$A_1 \rightarrow bA_3|bA_3A_4A_4|bA_4|bA_3A_4ZA_4|bZA_4$$

$$A_4 \rightarrow bA_3A_4|b|bA_3A_4Z|bZ$$

$$Z \rightarrow bA_3A_4A_4|bA_4|bA_3A_4ZA_4|bZA_4|bA_3A_4A_4Z|bA_4Z|bA_3A_4ZA_4Z|bZA_4Z$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$